SOC LAB 3 — SPLUNK SPL SEARCHES

Suspicious Office Document & LOLBin Execution — Searches separated from explanations

Purpose: Clean reference document for the supplied Splunk SPL searches, with each search followed by its explanation.

Search 1: Enterprise Office Document Triage

Screenshot / Panel Reference: 01_office_document_triage.png

Splunk SPL

```
index="soc-investigation" source="*office_doc*" EventCode="4688" ProcessName IN ("WINWORD.EXE", "EXCEL.EXE", "POWERPNT.EXE")
| dedup _time, ComputerName, ProcessId
| eval Delivery_Method=if(ParentProcessName="OUTLOOK.EXE", "Email Attachment (OUTLOOK.EXE)", "Local Disk (explorer.exe)")
| eval Risk_Verdict=case(
    like(CommandLine, "%.xlsm%"), "1. CRITICAL - Macro Email Attachment (.xlsm)",
    ParentProcessName="OUTLOOK.EXE", "2. Review - Standard Email Attachment",
    1=1, "3. Benign - Local Office File"
  )
| table _time, ComputerName, User, ParentProcessName, ProcessName, ProcessId, Delivery_Method, Risk_Verdict, CommandLine
| sort Risk_Verdict
```

Explanation

• `ProcessName IN ("WINWORD.EXE", "EXCEL.EXE", "POWERPNT.EXE")`: Retrieves all Microsoft Office application launches across all workstations.

• | eval Delivery_Method=if(ParentProcessName="OUTLOOK.EXE", ...): Distinguishes documents opened directly from Microsoft Outlook (`OUTLOOK.EXE`) vs. local files opened from `explorer.exe`.

• | eval Risk_Verdict=case(...): Flags the macro-enabled `.xlsm` attachment (`Q3_Tax_Adjustment.xlsm` opened by `EXCEL.EXE` PID 4800 on `FIN-WKSTN-04`) as `"1. CRITICAL"` and sorts it to Row 1.

Search 2: Office Macro & LOLBin Execution Triage

Screenshot / Panel Reference: 02_lolbin_child_processes.png & Dashboard Panel 4

Splunk SPL

```
index="soc-investigation" source="*office_doc*" EventCode="4688" (ProcessName IN ("cmd.exe", "certutil.exe", "rundll32.exe", "w
hoami.exe", "net.exe") OR ParentProcessName="EXCEL.EXE")
| dedup _time, ComputerName, ProcessId
| eval Verdict=if(ParentProcessName="explorer.exe" OR ProcessId IN (3295, 7310), "2. Benign Admin/User Command", "1. CRITICAL -
 Malicious Macro / LOLBin Execution")
| table _time, ComputerName, User, ParentProcessName, ProcessName, ProcessId, CommandLine, Verdict
| sort Verdict, _time
```

Explanation

• `(ProcessName IN ("cmd.exe", "certutil.exe", "rundll32.exe", "whoami.exe", "net.exe") OR ParentProcessName="EXCEL.EXE")`: Hunts for child processes spawned by `EXCEL.EXE` as well as Living-off-the-Land Binaries (LOLBins) and discovery utilities.

• | eval Verdict=if(...): Separates normal user commands launched from `explorer.exe` (`ping` and `ipconfig`) from the malicious macro chain on `FIN-WKSTN-04` (`EXCEL.EXE` ➔ `cmd.exe` PID 5200 ➔ `certutil.exe` PID 5650 and `rundll32.exe`).

Search 3: Correlated Incident Timeline on FIN-WKSTN-04

Screenshot / Panel Reference: 03_full_attack_chain_timeline.png & Dashboard Panel 5

Splunk SPL

```
index="soc-investigation" source="*office_doc*" ComputerName="FIN-WKSTN-04" "2023-11-16 10:14*" NOT ProcessName IN ("conhost.ex
e", "net1.exe")
| dedup _time, EventCode, ProcessId
| eval Attack_Stage=case(
    ProcessName="OUTLOOK.EXE" AND EventCode="11", "1. Phishing Attachment Cached (T1566.001)",
    ProcessName="EXCEL.EXE" AND EventCode="4688", "2. Malicious .xlsm Opened (T1204.002)",
    ProcessName="EXCEL.EXE" AND EventCode="11", "3. Excel Temp File Created",
    ProcessName="cmd.exe", "4. Macro Spawns cmd.exe (T1059.003)",
    ProcessName="certutil.exe" AND EventCode="4688", "5. LOLBin Downloader Launched (T1105)",
    ProcessName="certutil.exe" AND EventCode="3", "6. Outbound Payload Fetch (203.0.113.88:8080)",
    ProcessName="certutil.exe" AND EventCode="11", "7. Malicious DLL Dropped (calc_helper.dll)",
    ProcessName="rundll32.exe" AND EventCode="4688", "8. LOLBin DLL Execution (T1218.011)",
    ProcessName="rundll32.exe" AND EventCode="3", "9. Outbound C2 Channel (203.0.113.88:443)",
    ProcessName="whoami.exe", "10. Local Privilege Discovery (T1033)",
    ProcessName="net.exe", "11. Domain Account Discovery (T1087.002)",
    ProcessName="rundll32.exe" AND EventCode="11", "12. Startup Folder Persistence (T1547.001)"
  )
| eval Artifact=case(
    EventCode="11", TargetFilename,
    EventCode="3", DestinationIp . ":" . DestinationPort,
    EventCode="4688", CommandLine
  )
| table _time, Attack_Stage, EventCode, ParentProcessName, ProcessName, ProcessId, Artifact
| sort _time
```

Explanation

• `ComputerName="FIN-WKSTN-04" "2023-11-16 10:14*"`: Filters for all events on `FIN-WKSTN-04` during the `10:14` attack minute using wildcard string matching.

• | eval Attack_Stage=case(...): Labels each step of the kill chain from `1. Phishing Attachment Cached (T1566.001)` (`OUTLOOK.EXE` PID 2100) and `2. Malicious .xlsm Opened (T1204.002)` (`EXCEL.EXE` PID 4800) down to C2, discovery, and persistence.

• | eval Artifact=case(...): Displays `TargetFilename` for file creation events (`EventCode 11`), `DestinationIp:DestinationPort` for network connections (`EventCode 3`), and `CommandLine` for process creations (`EventCode 4688`).

Search 4: Lab 3 Dashboard Top-Row Chart Queries

Dashboard Panels: Panels 1, 2 & 3

Panel 1: Office Document Executions (by Risk Category) — Pie Chart

Splunk SPL

```
index="soc-investigation" source="*office_doc*" EventCode="4688" ProcessName IN ("WINWORD.EXE", "EXCEL.EXE", "POWERPNT.EXE")
| dedup _time, ComputerName, ProcessId
| eval Category=case(
    like(CommandLine, "%.xlsm%"), "Malicious Macro (.xlsm from Outlook)",
    ParentProcessName="OUTLOOK.EXE", "Benign Email Attachment (.docx/.xlsx)",
    1=1, "Benign Local Document (explorer.exe)"
  )
| stats count by Category
```

Panel 2: Office & LOLBin Process Ancestry (Parent → Child) — Bar Chart

Splunk SPL

```
index="soc-investigation" source="*office_doc*" EventCode="4688" ParentProcessName IN ("OUTLOOK.EXE", "EXCEL.EXE", "cmd.exe", "
rundll32.exe") NOT ProcessName IN ("conhost.exe", "net1.exe")
| dedup _time, ComputerName, ProcessId
| eval Process_Chain = ParentProcessName . " → " . ProcessName
| stats count by Process_Chain
| sort - count
```

Panel 3: FIN-WKSTN-04 Activity Over Time (By Event Type) — Column Chart

Splunk SPL

```
index="soc-investigation" source="*office_doc*" ComputerName="FIN-WKSTN-04"
| dedup _time, EventCode, ProcessId
| eval Event_Type=case(
    EventCode="4688", "Process Creation (4688)",
    EventCode="3", "Network Connection (3)",
    EventCode="11", "File Created (11)"
  )
| timechart span=5m count by Event_Type
```

Dashboard Explanation

• Panel 1 categorizes all Office executions by risk for the Pie Chart.

• Panel 2 concatenates `ParentProcessName` and `ProcessName` (`OUTLOOK.EXE → EXCEL.EXE`, `EXCEL.EXE → cmd.exe`, `cmd.exe → certutil.exe`, `cmd.exe → rundll32.exe`, `rundll32.exe → net.exe`, `rundll32.exe → whoami.exe`) for the Bar Chart.

• Panel 3 charts endpoint telemetry over time in 5-minute intervals.